Shraddha Jathar

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Summary

Data Scientist with 3+ years driving ML-powered solutions in fraud analytics, risk forecasting, and predictive maintenance. Expert in **Python**, **SQL**, **TensorFlow**, and **AWS** with demonstrated ability to reduce processing time by 30%, improve accuracy by 25%, automate critical workflows. Specialized in building production-grade ML pipelines, feature engineering, translating data science into business value.

Education

University of Texas at Arlington, Arlington, TX — *Aug 2023 – Aug 2025*
M.S. in Computer Science (Specialization: ML/AI and Database System) — GPA: 3.76

Skills

Programming: Python, SQL, R, VBA, Excel, Google Sheets, Tableau, Power BI, Scikit-learn, ELT Pipelines, LLM, LangChain.
Machine Learning & AI: TensorFlow, PyTorch, Keras, SVM, NLP (Sentiment Analysis, Text Mining), Computer Vision (OpenCV, Object Detection, Image Processing).
Cloud Platforms: AWS (Redshift, SageMaker, Lambda, EC2, S3), Azure, Apache Spark, Apache Airflow, Snowflake.
Data Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly.

Professional Experience

UT Arlington, Data Science Research Assistant — *May 2025 – Present*

- Conducted **data science-driven analysis** of inter-robot communication latency using **Python**, **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn**, applying **statistical analysis** and feature engineering to identify performance bottlenecks.
- Designed and executed ML-based experiments, including **reinforcement learning** driven positioning and **latency-aware** leader selection, to optimize coordination in autonomous surveillance systems.
- Collaborated with cross-functional research and engineering teams to evaluate AI-enabled and **generative AI (GenAI) assisted** simulation workflows, accelerating experiment design, data labeling, and result interpretation.

Mastercard, Data Scientist — *Aug 2024 – Apr 2025*

- Designed and deployed data pipelines for **fraud risk forecasting** and financial performance analysis using **Python** and **PySpark**, supporting planning, forecasting, business performance reporting, and improving data accuracy by 75% through optimized preprocessing techniques.
- Spearheaded **AI driven solutions** to optimize payment risk, transaction performance, and operational efficiency, enabling data-backed decision-making and reducing fraud processing time by 30%.
- Developed **predictive models** using **TensorFlow** and **PyTorch** to support risk stratification, P&L impact analysis, and chargeback reduction, contributing to a 25% decrease in chargeback incidents.
- Engineered and optimized **end-to-end ML pipelines** with **AWS SageMaker**, **Apache Spark**, and **Lambda**, automating data ingestion, feature extraction, and model deployment, reducing model training time by 40%.
- Leveraged **NLP techniques** to extract actionable insights from **unstructured payment data**, supporting business case development, pricing and risk analysis, and executive-level reporting.
- Collaborated with **cross-functional teams** including data scientists, software engineers, and financial analysts to design AI-powered tools that align with PCI DSS compliance and other regulatory requirements.
- Conducted **A/B testing**, variance analysis, and **model validation** to improve forecast reliability, performance measurement, and scenario analysis, achieving a 15% improvement in prediction robustness.

AT&T, Data Scientist — *Sep 2021 – Aug 2023*

- Built and deployed **machine learning models** using **Python** (**Scikit-learn, TensorFlow**), improving order fulfillment efficiency by 25% and enabling predictive insights for system anomalies and disruptions.
- Developed **ETL pipelines** using **Apache Airflow** and **Python**, automating data ingestion, transformation, and reporting, improving data quality and reducing manual errors by 40%.
- Optimized **SQL queries** for **large and complex datasets**, reducing query execution time by 30% and improving system performance for real-time data analysis and reporting.
- Created interactive dashboards in **Power BI** to visualize key business metrics, including order volume trends, failure rates, and resolution timelines, facilitating data-driven decision-making.
- Applied **time series models** for anomaly detection and **predictive modeling**, enhancing **forecasting accuracy** and early detection of operational disruptions, improving system reliability.

Academic Projects

LLM-Powered RAG Application (Python, LLamaIndex, OpenAI, Semantic Search, LangChain)

- Developed a production-ready **RAG** application leveraging LlamaIndex for document indexing and semantic retrieval paired with **OpenAI** API for generation. Compared multiple retrieval strategies (basic, sentence window, auto-merging) to optimize response quality and minimize hallucinations in document-based question-answering tasks.

Traffic Sign Recognition ML (Python, CNN, TensorFlow, OpenCV, NumPy)

- Built, tested, and optimized a **convolutional neural network** (CNN) in **TensorFlow**, achieving 95% accuracy on traffic sign classification. Developed an **ML pipeline** for data preprocessing and real-time inference, visualizing model performance and class-wise metrics with **Matplotlib** and **Seaborn**.